

PORT ISABEL HARBOR, TX, USA

WMO: 997369

Lat: 26.061N

Lon: 97.215W

Elev: 33

StdP: 14.68

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 40.2	(c) 44.0	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 26.9	(k) 62.7	(l) 24.8	(m) 61.1	(n) 10.4	(o) 330	(p) 0.469

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 6.8	(c) 89.3	(d) N/A	(e) 88.2	(f) N/A	(g) 87.3	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 12.1	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 25.3	(b) 23.3	(c) 20.8	(e) 37.6	(f) 93.2	(g) 4.1	(h) 1.7	(i) 34.6	(j) 94.4	(k) 32.3	(l) 95.4	(m) 30.0	(n) 96.3	(o) 27.0	(p) 97.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	74.2	60.8	63.3	68.3	74.3	78.8	82.7	83.3	84.2	81.7	78.1	69.5	64.7
	DBStd	9.85	8.03	7.88	6.56	4.83	3.56	1.72	1.89	1.56	3.00	4.88	7.92	7.93
	HDD50	34	18	7	1	0	0	0	0	0	0	0	4	4
	HDD65	498	171	114	47	7	0	0	0	0	0	2	51	106
	CDD50	8868	352	381	568	729	892	980	1034	1061	951	872	588	460
	CDD65	3855	40	67	149	285	427	530	569	596	501	409	184	98
	CDH74	37386	47	91	245	1364	3753	6273	6949	7620	5977	4035	813	219
	CDH80	11657	1	4	9	119	586	2081	2611	3199	2081	926	34	6
(14) Wind	WSAvg	10.6	8.9	10.8	12.5	13.3	12.3	11.4	11.2	9.8	8.8	9.7	9.5	9.7
(15) Precipitation	PrecAvg	25.1	1.0	1.2	1.3	1.6	2.0	2.3	1.9	1.9	5.3	3.5	1.8	1.3
	PrecMax	33.7	3.5	4.1	7.3	7.6	6.0	7.8	8.9	5.6	12.4	10.5	6.3	4.9
	PrecMin	14.3	0.0	0.0	0.1	0.0	0.0	0.3	0.1	0.0	1.0	0.0	0.0	0.1
	PrecStd	6.6	0.9	1.1	1.6	1.8	1.6	1.8	2.3	1.5	3.1	2.8	1.6	1.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.2	80.6	80.8	86.3	87.0	89.4	90.5	91.2	90.6	88.5	83.2	80.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	75.1	76.9	78.0	83.0	85.1	87.5	88.7	89.3	88.3	86.5	81.1	78.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	72.9	74.7	76.5	81.1	83.9	86.4	87.4	88.2	87.2	85.2	79.7	76.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	70.9	72.5	75.3	79.5	82.9	85.8	86.6	87.4	86.3	84.1	78.3	74.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	11.6	10.4	8.7	7.5	6.1	6.2	6.6	6.8	7.8	8.8	9.9	11.2
		MCDBR	12.5	11.8	8.3	9.2	6.6	7.1	7.2	7.7	7.8	8.4	9.1	10.9
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.372	0.384	0.402	0.453	0.470	0.438	0.457	0.442	0.444	0.397	0.381	0.375
	taud		2.467	2.425	2.373	2.226	2.210	2.371	2.323	2.376	2.335	2.465	2.470	2.459
	Ebn at Noon		279	283	284	270	264	271	266	270	267	275	273	272
	Edh at Noon		31	35	38	46	46	39	41	39	39	33	31	30
(44) All-Sky Solar Radiation	RadAvg		979	1189	1477	1763	1972	2115	2094	2008	1668	1463	1116	888
	RadStd		108	172	148	148	115	78	141	83	116	129	114	118

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+212

Nomenclature: See separate page